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Research

Intrusion detection in IoT networks using machine learning and deep learning approaches for MitM attack mitigation

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Abstract
The Internet of Things (IoT) revolutionizes connectivity but introduces significant security challenges, particularly concerning man-in-the-middle (MitM) attacks. These attacks, where adversaries intercept or alter communications, threaten the integrity and privacy of IoT networks due to the limited computational resources and security features of IoT devices. This study investigates machine learning and deep learning techniques to detect and mitigate MitM attacks in IoT environments. Three approaches, Long Short-Term Memory (LSTM), Random Forest, and Support Vector Machine (SVM), were assessed using a pre-prepared dataset that simulates network traffic and various attack scenarios. Random Forest achieved the highest accuracy (94%), combining robustness and balanced performance. LSTM excelled in analyzing sequential data with 92% accuracy, while SVM struggled with high false positive rates despite 85.7% accuracy. These findings highlight the potential of machine learning, particularly Random Forest and LSTM, to enhance IoT security. They pave the way for more resilient and effective frameworks for detecting and mitigating MitM attacks.

Article Highlights

1 The aggregate network of linked devices and the technology that enables communication between devices and the cloud, as well as between devices, are referred to as the Internet of Things, or IoT. We now have billions of devices connected to the internet because of the development of low-cost computer chips and high-bandwidth communications. This implies that commonplace appliances like vacuums, vehicles, machinery, and toothbrushes can employ sensors to gather information and react to consumers in a clever way.
2 A man-in-the-middle (MITM) attack is a type of cyberattack where a hacker intercepts communications between two internet targets, like a user and a web service, in order to obtain confidential data.
3 MITM attackers surreptitiously insert themselves into two-party interactions, then intercept private information including login credentials, account information, and credit card numbers. Hackers then utilize the data to carry out additional online crimes like identity theft, financial account takeover, and unlawful transactions.
4 An MITM attacker may intercept private conversations between two individuals in addition to user-application exchanges. In order to manage the conversation, the attacker in this scenario redirects and passes communications between the two parties, occasionally changing or replacing messages.
5 In order to identify and lessen MitM assaults in IoT contexts, this study explores machine learning and deep learning approaches. Using a pre-made dataset that mimics network traffic and different attack situations, three methods—Long Short-Term Memory (LSTM), Random Forest, and Support Vector Machine (SVM)—were evaluated. Random

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Forest combined robustness and balanced performance to attain the best accuracy of 94%. While SVM struggled with significant false positive rates despite 85.7% accuracy, LSTM performed exceptionally well in evaluating sequential data with 92% accuracy. These results demonstrate how machine learning, specifically Random Forest and LSTM, may improve IoT security. They open the door to frameworks for mitigating and detecting MitM assaults that are more robust and efficient.

Keywords IoT · MitM attack · Network security · Eavesdropping · Intrusion detection · Cybersecurity in IoT · Denial of the service · Machine learning · Deep learning

1 Introduction

The Internet of Things (IoT) represents a multifaceted and rapidly evolving paradigm that encompasses a network of interconnected devices capable of exchanging data autonomously, without the need for human-to-machine interaction. These devices span a wide spectrum, from simple sensors to sophisticated systems such as smartphones, smart home devices, and industrial automation tools. By integrating the physical environment with advanced hardware and software, IoT systems can make intelligent decisions based on real-time environmental inputs, significantly enhancing efficiency and functionality across various domains [4, 20]. The Internet of Things (IoT) aims to create a connected ecosystem of physical objects embedded with sensors, software, and other technologies, enabling them to collect and exchange data over the internet. This connectivity facilitates automation, improves efficiency, enhances decision-making processes, and provides new insights across various domains. This transformation has the potential to revolutionize various aspects of our lives, from everyday conveniences to large-scale industrial processes. Managing IoT becomes an enormous undertaking due to the heterogeneity of devices. Variations in power retention, data processing, memory capacity, bandwidth support, and more add complexity to the task.

Figure 1 highlight four main components of a typical IoT system categorized as: Smart things (sensors and actuators), connectivity, middleware/ data processing, and applications (user interface). Smart things like sensors collect data from the environment, while actuators perform actions based on received commands [8]. Connectivity ensures communication between devices and systems, utilizing various wired and wireless protocols. Middleware often comes first as it handles the initial integration and management of data before or as part of the data processing steps. Data processing can be embedded within the middleware or be a distinct process that follows the initial data handling by the middleware. Data processing, which can occur at the edge or in the cloud, analyzes the collected data, and the user interface allows users to visualize and interact with the system through dashboards and applications [9].

A man-in-the-middle (MitM) attack occurs when an unauthorized outsider intercepts, monitors, and potentially alters communication between two parties without their knowledge. This intrusion compromises the confidentiality, integrity, and availability of the data. Essentially, MitM attacks involve attackers intercepting traffic between two parties, usually without their awareness, to eavesdrop on or modify the communication. Figure 2 gives a graphical representation for MitM attack. MitM attack involves two main stages: interception and decryption. In a conventional MitM attack, the attacker exploits vulnerabilities in unsecured or poorly secured Wi-Fi networks, often found in public areas or homes, to intercept communication between a user's device and websites. The attacker scans the router for weaknesses, such as default passwords or poor configurations, and then inserts their tools between the user and the sites they visit [11].

A newer variation, known as a man-in-the-browser (MitB) attack, Fig. 3, requires the attacker to inject malware into the victim's computer. This malware embeds itself into the browser, recording and transmitting data exchanged between the user and specific targeted websites, like financial institutions, without the user's knowledge. Once the malware gathers the targeted information, it sends this data back to the attacker [23]. This malware is usually delivered through phishing emails, malicious downloads, or compromised websites. The malware integrates itself into the victim's web browser, allowing it to monitor and manipulate web traffic in real time without the user's knowledge.

This paper aims to address the challenges of analyzing network data for Internet of Things (IoT) environments and detecting attacks by utilizing machine learning and deep learning techniques, specifically Support Vector Machines (SVM), Random Forest, and Long Short-Term Memory (LSTM) networks. LSTMs, designed for sequential data, are particularly well-suited for IoT network traffic, which exhibits time-dependent patterns critical for accurate attack detection. The objectives include developing an automated intrusion detection system tailored for IoT networks, improving the accuracy of attack classification, and exploring feature extraction and reduction techniques to enhance detection performance.

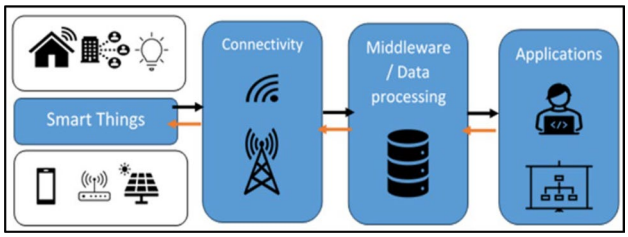

Fig. 1 The skeleton of an IoT system

The methodology involves training three models on IoT network traffic data annotated with attack types, optimizing the models to classify attacks based on temporal relationships in the data. Metrics such as accuracy and precision are used to evaluate performance. Unlike traditional methods or models relying on shallow machine learning techniques, this approach leverages LSTM's ability to capture long-term dependencies in sequential data. Furthermore, feature selection and reduction techniques are explored to improve model efficiency and accuracy, particularly for detecting complex attacks like Man-in-the-Middle (MitM), which are common in IoT environments.

The paper is organized as follows: Sect. 2 covers the study background, Sect. 3 reviews related work, Sect. 4 explains the methodology, Sect. 5 presents experimental results, Sect. 6 discusses the findings, and Sect. 7 concludes with future research directions.

2 Background

2.1 Techniques used in man-in-the-middle attacks in IoT

MitM attack can be executed in different communication channels such as GSM, UMTS, Long-Term Evolution (LTE), Bluetooth, Near Field Communication (NFC), and Wi-Fi. The attack targets are not only the actual data that flows between endpoints, but also the confidentiality and integrity of the data itself [6] MitM attacks can be conducted in several ways:

- Address Resolution Protocol (ARP) cache poisoning.
- DNS spoofing.
- Session hijacking including side-jacking, evil twin, sniffing.
- SSL Hijacking.

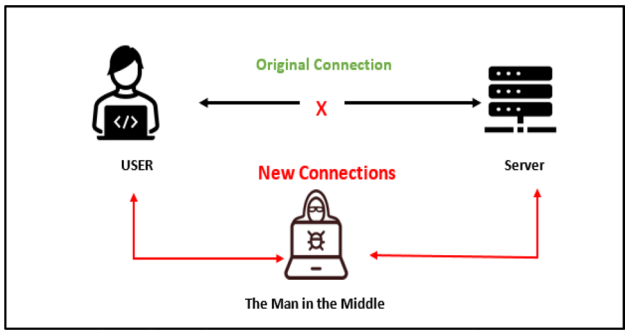

Fig. 2 Man in the middle attack

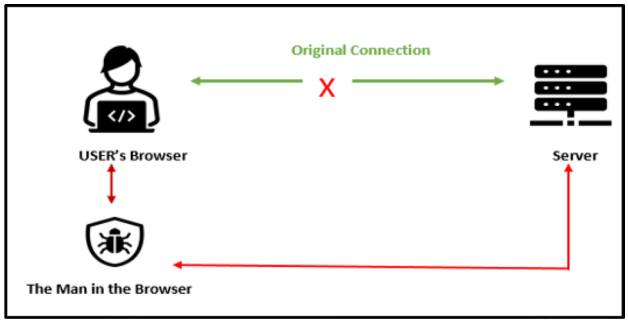

Fig. 3 Man in the browser attack

A commonly used model to assess the various aspects of a threat is the Confidentiality, Integrity, Availability (CIA) triad [17], however, MitM attacks directly affect all three sides of the triad, namely availability, integrity, and confidentiality, which pose significant risks and impacts on Internet of Things (IoT) systems.

According to Fereidouni et al. [7] it is possible to classify MitM attacks into two different categories, which are Passive attack and Active attacks.

- A passive MitM attack involves an attacker covertly listening to communication without altering it. By eavesdropping, the attacker can gain access to sensitive information such as usernames, passwords, and other confidential data, especially if the traffic is not properly encrypted.
- An active MitM attack occurs when an attacker intentionally intercepts and alters the communication between two parties. By manipulating the communication channel, the attacker can intercept, modify, or insert new messages. Common types of active MitM attacks include impersonation, spoofing, and data tampering.

Executing a MitM attack in an IoT environment typically involves three major steps, ranging from simple eavesdropping to complex data tampering [7]. The first step is reconnaissance, where the attacker gathers information about the target environment. In the second step, the attacker exploits vulnerabilities to intercept the communication between IoT devices. The final step involves modifying the intercepted data. Each of these steps can be facilitated by specific tools designed for the task. These steps could be conducted using respective tools.

i. Network scanning and targeting: To implement MitM attack the attacker scans the network to discover available devices, IP addresses, and open ports. Tools like Nmap [15] or Wireshark [5] are often used for this purpose. The attacker maps out the network topology to identify potential entry points.
ii. Interception and traffic analysis: The attacker uses techniques such as ARP spoofing, DNS spoofing, or setting up rogue access points to intercept data traffic. By tricking the network, the attacker positions themselves between the communicating parties, capturing the data packets being exchanged. Moreover, in IoT settings, lightweight brokers such as Mosquitto Broker ("Mallory,") are available as open-source MQTT brokers.
iii. Data Modification and vulnerability exploitation: The attacker alters the intercepted data before forwarding it to the intended recipient. This can involve injecting malicious code, modifying transaction details, or changing messages.

2.2 Challenges in detecting and preventing man-in-the-middle attacks

Detecting and preventing MitM attacks in IoT environments pose several significant challenges due to the unique characteristics of IoT devices and networks [6]. That, The MitM attack exploit the vulnerabilities and limited resources of sensors. Several publicly available tools (such as GATTacker, BtleJuice, Mirage, etc.) can be leveraged to launch MitM attacks [19]. Normally, IoT devices often have limited computational power, memory, and energy resources. This limits their ability to perform complex cryptographic operations or run intensive security monitoring and detection mechanisms. On the other side IoT devices come from diverse manufacturers, run different operating systems, and use various communication protocols. This heterogeneity complicates the implementation of standardized security measures and makes it challenging to enforce consistent security policies across all devices [13]. While encryption is crucial for securing communications, it adds computational overhead and latency, which can be problematic for resource-constrained IoT devices. Balancing security with device performance is a constant challenge [3].

3 Related work

A MitM attack is executed by attackers to intercept and capture sensitive information transmitted over a network. During the communication between two legitimate parties, the attacker covertly inserts themselves into the conversation without either party being aware. The attacker can eavesdrop, impersonate, or even alter the data being exchanged. Various studies have been conducted on this type of attack, and their findings are summarized below. These attacks can occur across different network types, Panda et al. [16] categorized MitM attacks on IoT networks and non-IoT networks attacks.

Wong and Luo [25] introduce a new scheme for a MitM attack on IoT devices that employ Message Queuing Telemetry Transport (MQTT) protocol, The proposed work presented the design of the attack, and conducted experiments using a

test-bed with IoT hardware and software, showing that MitM attack is effective against a wide range of typical anomaly detection mechanisms. Agyemang et al. [1] introduces a lightweight, real-time MitM detection and defense algorithm for Wi-Fi-enabled IoT gateways, addressing the inadequacies of current solutions for resource-constrained IoT systems. The algorithm relies on three sub-level processes: packet analyzer, detection and defense sub-process, and Asynchronous Method Dispatch (AMD) in inter-process communication.

Machine learning plays a significant role in detecting MitM attacks by leveraging its ability to analyze vast amounts of data and identify patterns indicative of malicious activity. Recently there have been a lot of research papers that explore the application of machine learning for secure utilization of IoT networks and devices. Sultan et al. [22] implement a machine learning model to classify MitM attacks on IoT devices using the MQTT protocol, highlighting the reliance on machine learning for enhancing IoT cybersecurity. Their work involves five different machine learning models, including Artificial Neural Network (ANN), Naïve Bayes (NB), eXtreme Gradient Boosting (XGB), Decision Tree (DT), and K-Nearest Neighbor (KNN), tested on an open-source dataset to classify MitM attacks and other security threats like DoS attacks.

Nwafor et al. [14] propose IoT-MGSec, a graph-based learning approach for detecting MitM attacks in IoT networks, which outperforms baseline models and offers a robust solution for enhancing IoT security. The model transforms network data into graphs to encapsulate intricate interactions, generating embedding to develop a robust classifier. [24] discuss detecting MitM attacks in 5G IoT networks using machine learning algorithms, with Gaussian Naive Bayes (GNB) demonstrating superior performance in prediction, testing time, precision, and recall compared to other methods.

Sivasankari and Kamalakkannan [21] propose a Regression Modeling technique to detect and mitigate MitM attacks in IoT networks, with Gaussian Process Regression showing higher detection rates and lower misclassification rates compared to Linear Regression and Multivariate Linear Regression. Michelena et al. [12] present an intelligent approach to detect MitM attacks on IoT using the MQTT protocol, involving the development of an intrusion detection system using feature extraction and supervised classification techniques. They created a new dataset capturing real traffic in an IoT environment with the MQTT protocol due to the absence of specific vulnerabilities in existing datasets.

The proposed works on detecting MitM attacks in IoT environments can be categorized based on their approaches and methodologies. While Wong and Luo [25] design the MitM attack and demonstrate the attack's effectiveness against typical anomaly detection mechanisms, other papers introduced different mechanism to detect MitM attack in IoT environment, Agyemang et al. [1] offer a lightweight, real-time detection and defense algorithm tailored for resource-constrained IoT gateways. Several studies emphasize the use of machine learning models for classification: Sultan et al. [22] highlight the utility of various classifiers in identifying MitM attacks, [24] focus on machine learning in 5G IoT networks with Gaussian Naive Bayes (GNB) showing superior performance, and Sivasankari and Kamalakkannan [21] utilize regression modeling techniques, particularly emphasizing Gaussian Process Regression.

Nwafor et al. [14] propose a novel graph-based learning approach, leveraging network data transformations for robust detection of MitM attacks. Lastly, Michelena et al. [12] introduce a hybrid intrusion detection system that combines feature extraction and supervised classification, using a new dataset to enhance detection capabilities. These varied approaches reflect the diverse strategies being developed to enhance the security of IoT environments against MitM attacks.

Table 1, summarizes the various proposed works introduced to mitigate the security threat of MitM attacks in IoT systems. It highlights the methods used, evaluation parameters, advantages, and potential future work for each approach.

4 Methodology

In this section, the work aims to develop a robust detection system capable of identifying MitM attacks in Internet of Things (IoT) environments. The approach leverages advanced machine learning and deep learning techniques due to their ability to learn and adapt effectively to complex patterns in data. Specifically, this research proposes a method for identifying MitM attacks using Support Vector Machines (SVM), Random Forest, and Long Short-Term Memory (LSTM) networks.

Below is a brief description of three specific techniques that can serve as effective mitigation strategies:

SVM is a supervised learning algorithm primarily used for classification and regression tasks. In IoT, SVM can mitigate security threats by detecting anomalies in network traffic or device behavior. SVM separates data points into distinct classes by constructing hyperplanes. This capability makes it highly effective in distinguishing between normal and malicious activity, even in high-dimensional datasets [18].

Random Forest is an ensemble learning method that uses multiple decision trees to improve predictive accuracy. In IoT systems, Random Forest can be applied for fault detection, resource optimization, and intrusion detection.

Table 1 Different techniques for the MiTM attack detection and mitigation in IoT

Authors	Technique	Method	Parameter	Advantage	Evaluation metrics	Future scope
[25]	MitM attack scheme on IoT using MQTT protocol	Attack design and testing	- False negatives and the impact of using BERT in improving detection rates	Demonstrate the vulnerability of MQTT to MitM attack	The ability of the attack scheme to evade detection by various ML-based anomaly detection models	Exploring alternative MitM attack beyond using a Wi-Fi Pineapple and examining the application of BERT in IDS systems in IoT devices
[1]	Real-time MitM detection and defense algorithm designed for Wi-Fi-enabled IoT gateways	MitM detection and defense algorithm	- CPU utilization efficiency, detection rate and network latency (RTT)	Lightweight MitM detection and defense mechanism can for Wi-Fi-enabled IoT gateway devices	Performance overhead reduction	Enhance by improving efficiency, effectiveness, scalability, and adaptability
[22]	ML models to classify MitM attacks on IoT	- ANN - NB - XGB - Decision Tree (DT) - KNN	- Accuracy, precision, recall, F1 score - Training time and test time	- Improve cybersecurity in IoT networks	The accuracy, precision, Recall and F1 score on detect MitM attach in IoT environment Compare different ML algorithms(ANN),(NB),(XGB) and (DT)	Not specified
[14]	Graph-Based Learning Technique for Classifying MitM Attacks on IoT	- Network Graph - node2vec	- Precision, recall, and F-1 score	- More robust solution to detecting and mitigating MitM attack - Integration of real-time IDS systems	Precision, Recall and F1 score to evaluate the performance of proposed graph-based learning approach compared to a baseline model	Employing advanced graph learning models as Graph Convolution Networks and GraphSAGE
Sivasankari and Kamalakkannan [21]	Regression Modelling for MitM attack detection and mitigation	- Linear Regression (LR) - Multi-variate Linear Regression (MLR) - Gaussian Process Regression (GPR)	- Regression analysis - Performance of LR, MLR, and GPR techniques - Packet loss ratio, and throughput	Gaussian Process Regression, which proves to be more efficient in detecting MitM attacks in an IoT network	The rate of detecting attacks and the rate of misclassifying attacks	Developing advanced ML and real-time monitoring and response systems
[12]	MitM Attack Detection Using Supervised Classification and Principal Component Analysis	- Hybrid feature extraction and supervised classification techniques	- Principal component, Variance percentage - Performance of: RF - Area Under the receiving operating characteristic Curve (AUC)	Detection of MitM attacks in IoT using an IDS systems with intelligent models and advanced classification techniques	- Accuracy, Precision, Recall,F1-Score,Specificty,AUC,Training Time and Testing Time - Statistical methods: Kruskal–Wallis analysis of variance and Tukey's comparison	Exploring other types of dimensional reduction and feature extraction methods - Application of unsupervised (one-class) methods for future research

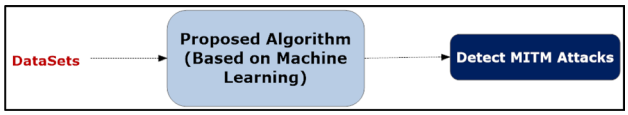

Fig. 4 General methodology

Fig. 5 Sample of data in the dataset

Type	inter_arriv	seq	duration	ttl	datarate	dbm_ants
Attack2	0	1684.0000	44	44	54	-36
Attack2	0.08759	1688.0000	44	44	54	-36
Attack2	0.12796	1689.0000	44	44	54	-34

Its capability to handle large datasets with diverse features makes it effective in analyzing heterogeneous IoT data. Additionally, its resilience to noise and overfitting ensures reliable performance in dynamic environments [2].

LSTM is a type of recurrent neural network (RNN) designed to process sequential data. In IoT networks, LSTM is particularly useful for time-series analysis, such as predicting network traffic patterns or identifying abnormal behavior over time [10].

The foundation of this approach lies in machine learning theory. Instead of building a virtual or real network environment for experimentation, the methodology utilizes a Pre-prepared dataset for wireless networks. This dataset, which includes diverse network scenarios and instances of MitM attacks, serves as a reliable source for training and evaluating the proposed models. Using such a dataset ensures that the models are trained on realistic data, enhancing their generalizability and effectiveness in detecting attacks. The pre-prepared dataset is used to implement the proposed detection algorithms. The general methodology is illustrated in Fig. 4. This dataset undergoes preprocessing to extract relevant features, ensuring the models are trained on high-quality inputs. The extracted features are then fed into machine learning and deep learning models. LSTM, in particular, is utilized to capture temporal dependencies in network traffic, while SVM and Random Forest provide complementary methods for comparative analysis.

A sample of this dataset that includes various network scenarios and instances of MitM attacks, is illustrated in Fig. 5.

To verify the quality of the proposed MitM detection algorithm, the methodology includes evaluation using test data. The primary evaluation metric is the ratio of correctly predicted attack cases to the total number of attacks. Additionally, the false prediction of attack cases contributes to assessing the algorithm's performance.

The proposed detection process is detailed in Fig. 6. The first step involves training the model using deep learning techniques. The data undergoes preprocessing and feature extraction to isolate critical features that improve the accuracy and efficiency of the model. The LSTM network is trained on this processed data, ensuring the generated model is both accurate and robust. Once trained, the model is applied to real Wi-Fi network data to classify and detect MitM attacks. The experiments utilize multiple datasets to reflect realistic wireless network behavior and the types of attacks they encounter.

Three experiments are conducted using SVM, Random Forest, and LSTM to compare the performance of these techniques. The Confusion Matrix, as illustrated in Fig. 7, serves as a key tool to evaluate the models. It helps calculate true positives, true negatives, false positives, and false negatives.

Additionally, evaluation metrics such as precision, recall, and the F1 score are computed using the following equations:

$$\text{Precision} = \frac{True\ Positive}{True\ Positive + False\ Positive}$$

$$\text{Recall} = \frac{True\ Positive}{True\ Positive + False\ Negative}$$

Fig. 6 Proposed algorithm

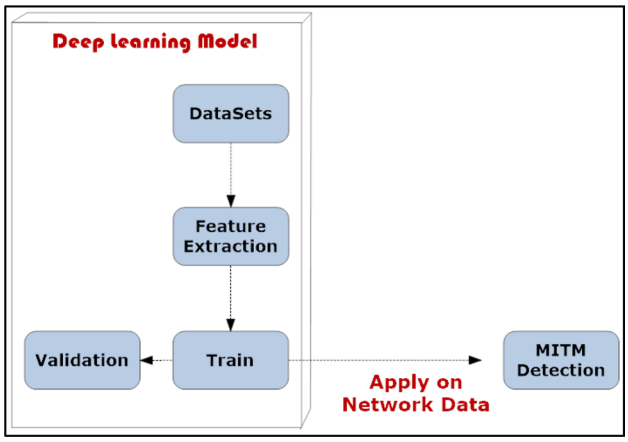

Fig. 7 Confusion matrix

		Predicted	
		Negative	**Positive**
Actual	**Negative**	True Negative	False Positive
	Positive	False Positive	True Negative

$$F1 = 2 \times \frac{Precision * Recall}{Precision + Recall}$$

5 Experiments results

The dataset used for this study is categorized into two classes: MitM attack and clean (non-attack) traffic. The dataset, named Normal—Attack1-Attack2, is labeled numerically for ease of training and processing. Specifically, clean traffic is assigned the label Normal = 0, while both attack types, Attack1 and Attack2, are labeled as 1. Figure 8 provides a visual representation of the distribution of these classes within the dataset. This distribution is critical to understanding the dataset's composition and ensuring that the model can learn effectively from the available data.

5.1 Attack detection based on deep learning technique (LSTM)

The LSTM (Long Short-Term Memory) model was trained using the previously prepared dataset, which includes both normal network traffic and instances of MitM attacks. The training process is depicted in Fig. 9, showcasing the training function used to optimize the model.

The performance of the trained model is further analyzed using the confusion matrix, as shown in Fig. 10, which highlights the distribution of predictions across the different classes (Normal and Attack).

The effectiveness of the LSTM model is evaluated using several metrics. The accuracy of the model is computed as 0.92, indicating a high overall correctness in predictions. The following are the detailed results from the confusion matrix:

- True positive (TP) = 454
- False positive (FP) = 426
- False negative (FN) = 97
- True negative (TN) = 5138

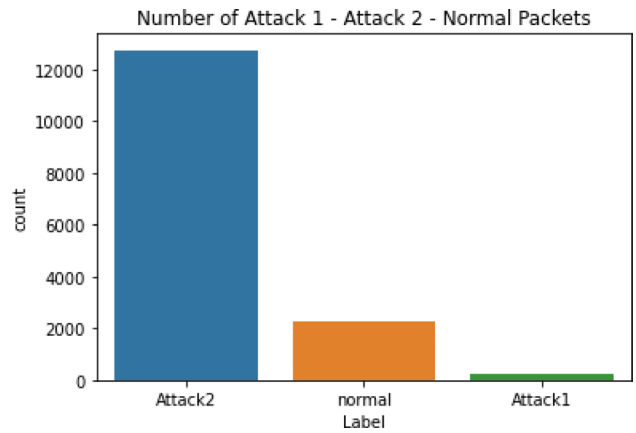

Fig. 8 Dataset distribution

Fig. 9 Training and validation loss

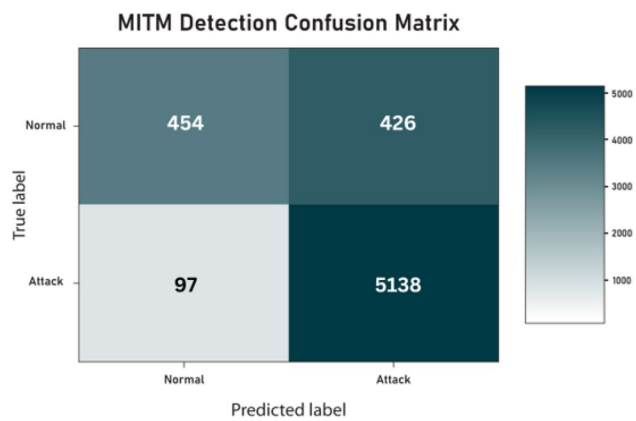

Fig. 10 Deep learning technique confusion matrix

Key performance metrics derived from these values include:

- Recall = 0.823
- Precision = 0.495
- F1 Score = 0.573

5.2 Attack identification using the Random Forest method

The Random Forest algorithm, a supervised learning technique, is utilized for attack detection in this study. As depicted in Fig. 11, Random Forest consists of an ensemble of decision tree classifiers. Each tree is trained on a

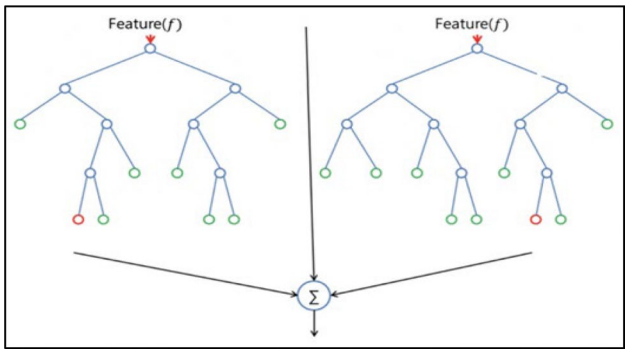

Fig. 11 Random Forest structure

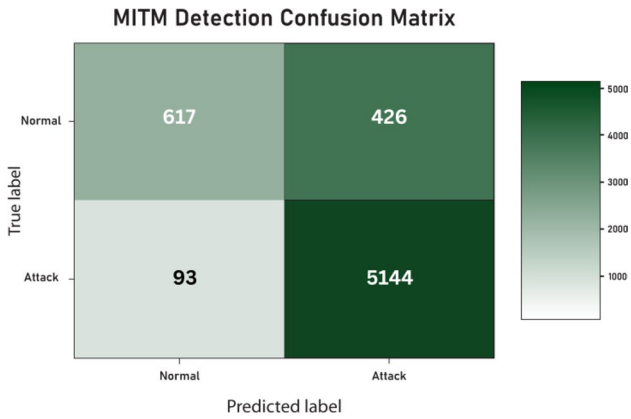

Fig. 12 Confusion matrix for Random Forest technique

subset of the data, and the final prediction is determined by aggregating the results from all individual trees. This approach improves the model's overall accuracy and reduces the risk of overfitting.

The performance of the Random Forest model is evaluated using the confusion matrix shown in Fig. 12.

The accuracy of this method is 0.94, demonstrating strong performance in identifying both normal and attack data. The following metrics are derived from the confusion matrix:

- True positive (TP) = 614
- False positive (FP) = 265
- False negative (FN) = 96
- True negative (TN) = 5141

Key performance metrics derived from these values include:

- Recall = 0.864
- Precision = 0.698
- F1 Score = 0.772

5.3 Attack detection based on SVM technique

The Support Vector Machine (SVM) is a binary classification technique that separates data into two classes using a hyperplane, as depicted in Fig. 13.

In this study, the SVM model is trained to classify network traffic as either normal or an attack. The results from the SVM classifier are summarized in the confusion matrix shown in Fig. 14.

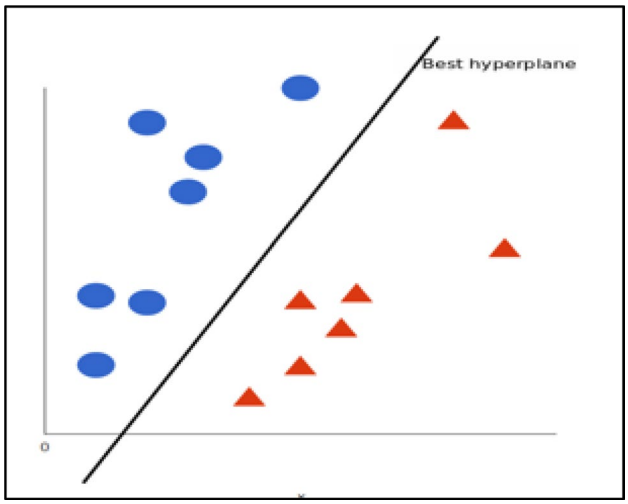

Fig. 13 Hyperplane by SVM

The accuracy of the SVM classifier is calculated at 0.857, but it is important to note that this relatively high accuracy is influenced by the misclassification of normal data as attacks, resulting in a higher number of false positives. The following confusion matrix values are observed:

- True positive (TP) = 0
- False positive (FP) = 876
- False negative (FN) = 0
- True negative (TN) = 5237

Key performance metrics derived from these values include:

- Recall = 0
- Precision = 0
- F1 Score = 0

While the SVM technique demonstrates an acceptable accuracy rate, its performance in detecting attacks is poor due to the high false positive rate and the inability to identify any actual attacks. This highlights the challenges of using SVM for attack detection in this context.

Finally, a comparison of the accuracies achieved by the three techniques (LSTM, Random Forest, and SVM) is illustrated in Fig. 15.

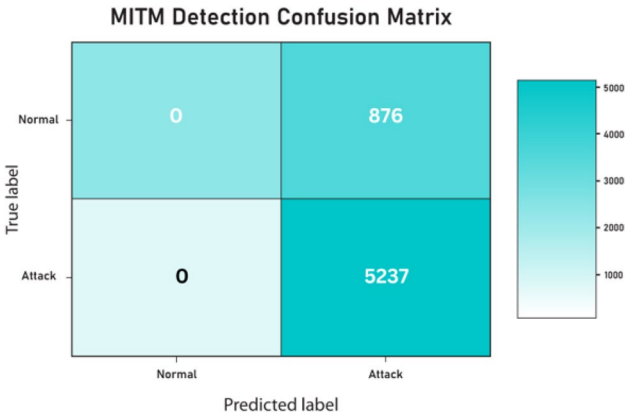

Fig. 14 Confusion matrix for SVM

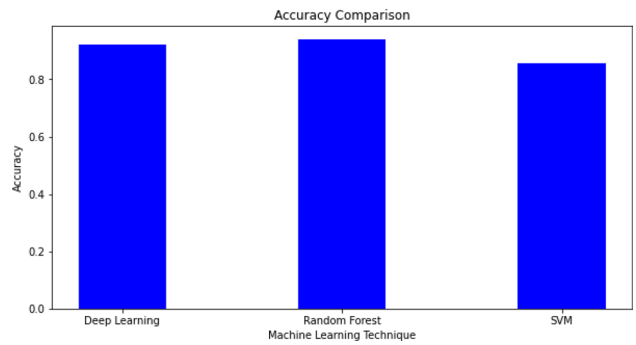

Fig. 15 Comparison of accuracy across the LSTM, Random Forest, and SVM techniques in attack detection

6 Discussion

The results of this study highlight the strengths and limitations of three different techniques—LSTM, Random Forest, and SVM—for detecting MitM attacks in IoT network environments.

The LSTM model, designed to handle sequential data, demonstrated a strong capability to analyze IoT network traffic, achieving an accuracy of 0.92. Its ability to capture temporal patterns allowed for effective attack detection, with a recall of 0.823. However, the precision (0.495) and F1 score (0.573) indicate room for improvement, particularly in reducing false positives to enhance reliability.

The Random Forest approach outperformed the other techniques, achieving the highest accuracy of 0.94. With a recall of 0.864, a precision of 0.698, and an F1 score of 0.772, it provided a balanced performance in detecting attacks while minimizing false positives. The ensemble nature of Random Forest, which aggregates decisions from multiple trees, contributed to its robustness and reliability.

In contrast, the SVM classifier struggled to identify attacks, achieving an accuracy of 0.857, but failing to detect any true positives. The high false positive rate and resulting precision, recall, and F1 scores of 0 highlight the limitations of SVM in this context. This result suggests that SVM is not well-suited for detecting complex attack patterns in IoT network traffic.

7 Conclusion

This study investigated the effectiveness of machine learning and deep learning techniques in detecting Man-in-the-Middle (MitM) attacks within Internet of Things (IoT) environments. By evaluating Long Short-Term Memory (LSTM), Random Forest, and Support Vector Machine (SVM) classifiers on a benchmark dataset, the study aimed to address the challenges associated with securing IoT networks against MitM attacks. The results demonstrated that Random Forest achieved the highest accuracy (94%) and offered a balanced performance in terms of recall (0.864) and precision (0.698). LSTM also showed promise with its ability to analyze sequential data, achieving an accuracy of 92%, though its precision (0.495) highlights areas for improvement. SVM, while achieving 85.7% accuracy, struggled to detect attacks effectively due to high false positive rates, indicating its limitations in handling complex IoT attack patterns. The findings underscore the potential of machine learning, particularly ensemble methods like Random Forest and deep learning techniques like LSTM, in enhancing IoT security.

Future research will focus on refining these models, exploring hybrid techniques that combine the strengths of Random Forest and LSTM, and addressing the resource constraints of IoT devices to implement real-time attack detection systems.

Author contributions M.A. and S.A.S. wrote the main manuscript text and prepared figure and table. All authors reviewed the manuscript.

Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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References

1. Agyemang JO, Kponyo JJ, Acquah I. Lightweight man-in-the-middle (MITM) detection and defense algorithm for WiFi-enabled Internet of Things (IoT) gateways. Inf Secur Comput Fraud. 2019;7(1):1–6.
2. Alahmadi AA, Aljabri M, Alhaidari F, Alharthi DJ, Rayani GE, Marghalani LA, et al. DDoS attack detection in IoT-based networks using machine learning models: a survey and research directions. Electronics. 2023;12(14):3103.
3. Alnasser S, Alsaqour R. Encryption algorithms modeling in detecting man in the middle attack in medical organizations. Int J Adv Comput Sci Appl. 2022. https://doi.org/10.14569/IJACSA.2022.0130810.
4. Berte DR. Defining the iot. In: Proceedings of the international conference on business excellence (Vol.12, No. 1, pp. 118–128). 2018.
5. Bullock J, Parker JT. Wireshark for security professionals: using wireshark and the metasploit framework. New Jersey: Wiley; 2017.
6. Conti M, Dragoni N, Lesyk V. A survey of man in the middle attacks. IEEE Commun Surv Tutor. 2016;18(3):2027–51.
7. Fereidouni H, Fadeitcheva O, Zalai M. IoT and man-in-the-middle attacks. arXiv preprint arXiv:2308.02479. 2023.
8. Fortino G, Trunfio P. Internet of things based on smart objects: technology, middleware and applications. Cham: Springer; 2014.
9. Greengard S. The internet of things. Cambridge: MIT Press; 2021.
10. Inuwa MM, Das R. A comparative analysis of various machine learning methods for anomaly detection in cyber attacks on IoT networks. Internet Things. 2024;26: 101162.
11. Mallik A. Man-in-the-middle-attack: Understanding in simple words. Cyberspace Jurnal Pendidikan Teknologi Informasi. 2019;2(2):109–34.
12. Michelena A, Aveleira-Mata J, Jove E, Bayón-Gutiérrez M, Novais P, Romero OF, et al. A novel intelligent approach for man-in-the-middle attacks detection over internet of things environments based on message queuing telemetry transport. Expert Syst. 2024;41(2): e13263.
13. Mishra S, Paul A. A critical analysis of attack detection schemes in IoT and open challenges. Paper presented at the 2020 IEEE international conference on computing, power and communication technologies (GUCON), 2020; 57–62.
14. Nwafor E, Schmidt C, Olufowobi H. IoT-MGSec: mitigating man-in-the-middle attacks in IoT networks using graph-based learning. Paper presented at the 2023 International Conference on Machine Learning and Applications (ICMLA), 1261–1266. 2023.
15. Orebaugh A, Pinkard B. Nmap in the enterprise: your guide to network scanning. Amsterdam: Elsevier; 2011.
16. Panda D, Mishra BK, Sharma K. A taxonomy on man-in-the-middle attack in IoT network. Paper presented at the 2022 4th International Conference on Advances in Computing, Communication Control and Networking (ICAC3N), 1907–1912. 2022.
17. Reshan A, Saleh M. IoT-based application of information security triad. Int J Interact Mobile Technol (iJIM). 2021;15(24):61–76. https://doi.org/10.3991/ijim.v15i24.27333.
18. Saleem AD, Abdulrahman AA. Attacks detection in Internet of Things using machine learning techniques: a review. J Appl Eng Technol Sci. 2024;6(1):684–703.
19. Salem O, Alsubhi K, Shaafi A, Gheryani M, Mehaoua A, Boutaba R. Man-in-the-middle attack mitigation in internet of medical things. IEEE Trans Industr Inf. 2021;18(3):2053–62.
20. Silverio-Fernández M, Renukappa S, Suresh S. What is a smart device? - A conceptualisation within the paradigm of the Internet of Things. Vis Eng. 2018. https://doi.org/10.1186/s40327-018-0063-8.
21. Sivasankari N, Kamalakkannan S. Detection and prevention of man-in-the-middle attack in IOT network using regression modeling. Adv Eng Softw. 2022;169: 103126.
22. Sultan ABM, Mehmood S, Zahid H. Man in the middle attack detection for MQTT based IOT devices using different machine learning algorithms. Paper presented at the 2022 2nd International Conference on Artificial Intelligence (ICAI), 118–121. 2022.
23. Tommasi F, Catalano C, Taurino I. Browser-in-the-middle (BitM) attack. Int J Inf Secur. 2022;21(2):179–89.
24. Vijayalakshmi A, Broody J, Sathishkumar JS. Detection of man in the middle attack in 5G IOT using machine learning. Paper presented at the 2023 International Conference on Recent Advances in Science and Engineering Technology (ICRASET), 1–5. 2023.
25. Wong H, Luo T. Man-in-the-middle attacks on MQTT-based IOT using Bert based adversarial message generation. Paper presented at the KDD 2020 AIoT Workshop. 2020

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